

COMP20008 Elements of Data Processing



Lecture 1: Introduction to the Subject and Data Processing

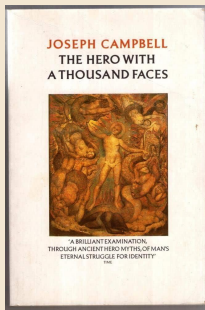
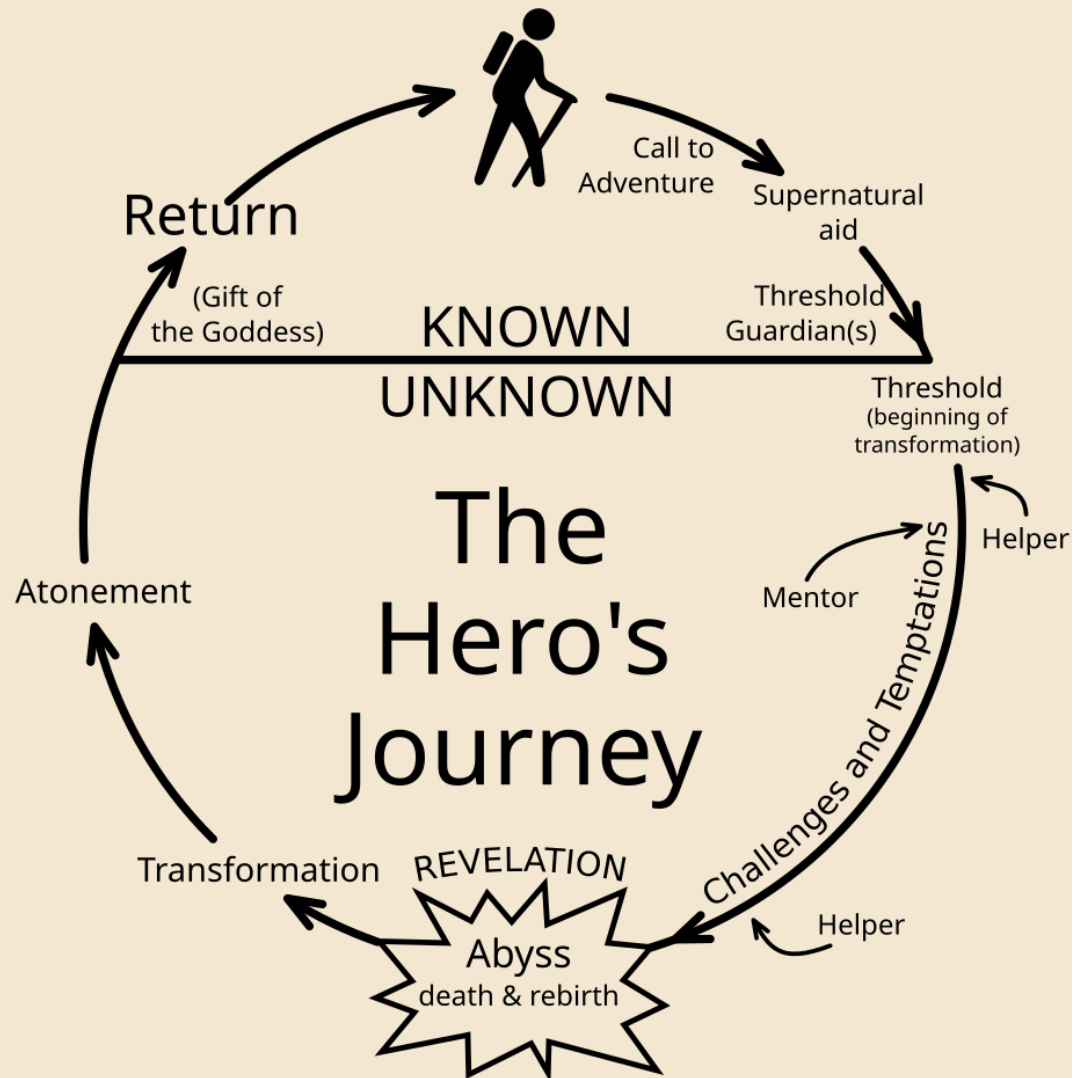
Semester 2 2026

Subject Coordinator: **Dr Ekaterina Vylomova**

Co-lecturer: **Dr Hanzhuo (Vivian) Ma**



Welcome to the Journey in the DataLand!



Helpers: Lecturers

Subject Coordinator

Dr Ekaterina
Vylomova

Research: technology for Indigenous languages

Fulbright Scholar (2014), Google PhD Fellow (2017), ARC DECRA Fellow (2026–2028), Excellence Award in EC Research (2026)



Co-lecturer

Dr Hanzhuo (Vivian)
Ma

Research: management information systems

Visiting Scholar (Indiana University, 2023); Best Doctoral Thesis Award (2026)



Helpers: Teaching Team

Head Tutor

- Hasti Samadi



Teaching Staff

- Reyhaneh Goli



Tutors

- Andrew Pham
- Di Wu
- Farnaz Pirasteh
- Mitra Heidari
- Kenny Lam
- Lenny Bloom

Markers

- Vincent Barbosa
- Navid Akhavan Attar
- Tifani Bo
- Yuchen (Steven) Luo
- Kenzi Lamberto
- Wen Zhou

The goal of the subject

After this subject you should know:

- **The major forms of data** (various types of structured, semi-structured, and unstructured data)
- **The major formats of data** (how the data is actually represented inside the computer)
- What sorts of **operations** you typically (can) do on the **various forms and formats**
- **Ways of checking** that your computing is correct
- Some **common models, algorithms, and Python packages** you can use for data processing

The Journey



Data Input and Formats:

Data Types:
structured/unstructured;
continuous/discrete;
numerical/text

Data Formats:
JSON, CSV, XML, HTML

Data collection:
crawling, scraping

Text processing

LLMs for data processing:
prompt engineering & eval;
RAG-based systems

The Journey



Data Cleanup and Preparation:

Data cleanup, anomaly and outlier detection

Preparation: scaling, imputation, sampling, data extension

The Journey



Data Interpretation:

Clustering: k-means, VAT, hierarchical

Dimensionality reduction: PCA

Correlation analysis, entropy, mutual information

Classification: k-NN, decision trees

Regression

Experimental design and machine learning

The Journey



Data Visualisation:

Clustering: k-means, VAT, hierarchical

Dimensionality reduction: PCA

Approaches to visualise statistical characteristics, histograms, heatmaps

Dynamic visualisation

The Journey



Data Use and Ethics:

What is Ethics?

Data Bias

Privacy: self privacy, local privacy, global privacy

Intellectual Property

GenAI

Weekly topics – Draft schedule

- Week 1: Course Overview and Data representation
- Week 2: Structured Data: Data Capture and pre-processing
- Week 3: Unstructured Data: Text Capture and Processing
- Week 4: Document Processing and Visualisation
- Week 5: Data Applications I: Clustering and PCA
- Week 6: Data Applications II: Correlation
- Week 7: Data Applications III: Supervised learning
- Week 8: Data Applications IV: Experimental design and evaluation
- Week 9: AI Ethics & Intellectual Property & Privacy
- Mid-semester Break
- Week 10: Prompt Engineering for LLMs and Oral Presentation training
- Week 11: Generative AI and Large Language Models (LLMs)
- Week 12: Guest Lecture and Conclusion

Background knowledge needed

1. **Python programming language**
2. **Mathematical skills**
 - Mathematical notation (logs, powers, distances,...)
 - Familiar with basic probability and statistics (mean, median, etc)
 - Information theory (a plus!)
3. **Subject is at 2nd year level**
 - More independence
 - Will need to be comfortable looking up documentation for Python library functions

What's not covered?

- Advanced predictive analytics, machine learning
 - COMP30027 Machine learning
- Relational databases
 - INFO20003 Database Systems

What the subject is:

- What the subject is:
 - A broad coverage of the steps involved in the data processing pipeline
- What the subject is not:
 - A deep dive into any one particular area of the pipeline
 - A programming subject

Subject structure

- **Lectures and Workshops:**
 - Two lectures every week, 1 hour each: **Mondays at 5pm; Thursdays at 12pm**
- One workshop every week, 2 hours
 - **Workshops start in Week 1**
 - Opportunity to practice data science/wrangling techniques
 - Please come prepared
 - **Bonus marks for attendance and engagement!**

Subject structure

- **Assignments:**

- 2 assignments, due dates throughout semester.
See draft schedule on Canvas
(Assignment 1 - Individual, Assignment 2 - Group)
- For Assignment 2 groups should be formed by Week 5!
- Marks count 50% toward final mark
 - Hurdle 20/50

- **Final exam:**

- Final written exam during the examination period
- Score counts 50% toward final mark
 - Hurdle 20/50

Oral Assessment



Week 8

Presented remotely via
Zoom



Week 12

Presented in-person

Academic Integrity

- Make sure the work you submit is yours
- Never share any examinable code with students outside your project group (not on the forums, email, via shared devices, ...)
- Review the student support modules on the COMP20008 canvas page on academic integrity

Resources

- **Canvas**
 - Announcements
 - Lecture materials
 - Quizzes (Practice, released weekly, starting from Week 2)
- **Ed Lessons**
 - Jupyter Notebooks for workshops
- **Ed discussion board**
 - Peer-based discussions
 - Tutors and lecturers answering questions (workshops and lectures)
- **Assignment consultations**
 - Extra consultation sessions will be scheduled during assignment due date periods.

Announcement 1 – Student Rep



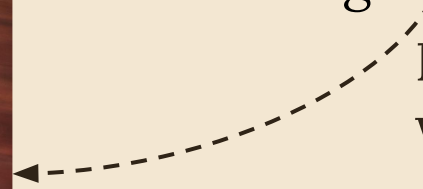
If you are interested, please contact Dr Ekaterina Vylomova

Announcement 2 - Group



Find your group
members
for assignment 2!

By the end of
week 5!



FIN!